

Steve Gwade

ML Engineer (Applied Research)

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Professional Summary

ML Engineer with 3 years building and deploying ML systems across healthcare, agriculture, finance, transport, and consumer products. Skilled in RAG systems, real-time inference, computer vision, automation, and applied research with measurable production impact.

Experience

- **Lead Backend Engineer (Freelance)**, MITS SARL (*Mar 2026–May 2026*) – engineered an LLM-driven conversational booking workflow with bilingual interaction and concurrency-safe approval logic, reducing booking request time by over 90%.
- **Research Engineer (ML & Full-Stack)**, JIANTS (*Jan 2025–Jan 2026*) – conducted internal experimentation for vision, OCR, STT, and TTS use cases, reducing prototype-to-deployment time by about 30%, and contributed to LINA, improving navigation and object recognition by 56%.
- **Software Engineer (AI & Full Stack)**, TGNix (*Mar 2024–Dec 2024*) – built AI-backed systems and integrated ML inference into production applications, improving workflow speed by 25%.
- **Junior Backend Engineer (ML & Computer Vision)**, AITECAF (*Jun 2023–Nov 2023*) – built APIs for AI services, developed LangChain assistants, and trained a YOLO cassava disease model enabling 98% accuracy and sub-5ms inference.

Selected Projects

- **VERO** (Personal, Done) Grounded research workspace with project-scoped RAG, hybrid retrieval, and citation-aware multi-turn answering over documents, URLs, and repos. Improved and helped accelerate literature review by over 44%.
- **PlumVision** (JCIA Hackathon Winner, 2025) Real-time plum classifier with ConvNeXt and contrastive learning, 98% accuracy, 3MB model, sub-12ms FastAPI inference.
- **EatWise** (JCIA Most Innovative AI Prototype, Ongoing) Food recognition with object detection, personalized nutrition and meal recommendations, real-time FastAPI inference.
- **Spoof Detector** (Personal) Deep CNN face spoof detection with Flask API for secure authentication integration.
- **Multi-Task Transfer Learning for Malaria Diagnosis** (MSc Thesis, 2025) Joint multi-task model achieving 97% macro F1, deployed via FastAPI, outperforming single-task baselines.

Awards and Recognition

- **2nd Place Orange Digital Centre Champions Africa 2025**, algorithmic coding contest across Africa.
- **JCIA Hackathon 2025 Winner**, *PlumVision* – 98% accuracy, 3MB model, <12ms inference, instant quality control.
- **Most Innovative AI Prototype JCIA AI Solutions Track Winner**, *EatWise* – personalized nutrition assistant.
- **1st Prize Huawei ICT Competition**, *LINA* – AI system improving navigation, and object recognition for blind users by 56%.

Academic Qualifications

- **MSc in Engineering** (*2023–2025*), National Higher Polytechnic School of Douala – Data Science & AI, thesis: multi-task transfer learning for malaria.
- **BSc in Engineering** (*2020–2023*), National Higher Polytechnic School of Douala – Computer Science and foundations of DS/AI.

Research & Publications

- **Multi-Task Transfer Learning and Application to Malaria Detection** – MSc Thesis, 2025, github.com/GwadeSteve/MTTL-Research-Malaria.

Skills & Interests

Technical: Python, JavaScript, SQL, PyTorch, TensorFlow, Keras, Jax, scikit-learn, LangChain, RAG, Docker, CI/CD, GitHub Actions, MLOps, MLflow, Weights & Biases, AWS (EC2, Fargate, S3, SageMaker), REST APIs, PostgreSQL, MongoDB, Linux.

Machine Learning: Supervised, Self-/Unsupervised, Representation Learning, Transfer Learning, Data Engineering, Object Detection, Image Classification, Segmentation, Feature Extraction, Contrastive Learning, Embeddings, Real-time Inference Pipelines.

Professional: Problem solving, End-to-end ML project execution, Experiment design, Critical thinking, Stakeholder communication, Translating business needs into AI solutions, Ownership, Efficient delivery.

Interests: Reading, Gaming, Basketball, Coffee, Open-source contribution.